

# Yuhao Zhu

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## CONTACT INFORMATION

Associate Professor  
University of Rochester  
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Rochester, NY 14627

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<https://yuhaozhu.com/>  
<https://horizon-lab.org/>  
<https://github.com/horizon-research>

## RESEARCH INTERESTS

I do research in computer systems (computer architecture, compilers, and programming models). My current research aims to develop the engineering foundations for embodied intelligence (e.g., Extended Reality, smart wearables, autonomous agents/robots) by co-designing systems across three boundaries traditionally treated separately: 1) the sensing-computing boundary through which machines model the world, 2) the software-hardware boundary through which information in the physical world is efficiently interpreted, and 3) the computing-human boundary where human factors determine what should be computed and how, which in turn shape human perception, cognition, and action.

## EDUCATION

### The University of Texas at Austin

Ph.D., Electrical and Computer Engineering

*May 2017*

M.S.E., Electrical and Computer Engineering

*May 2015*

Dissertation: [Energy-Efficient Mobile Web Computing](#)

### Beihang University, Beijing, China

B.S., Computer Science and Engineering

*June 2010*

## ACADEMIC HONORS

- 2026-2029, [Asaro Biggar Family Fellow in Data Science](#), University of Rochester
- 2023-2024, [Bridging Fellowship](#), University of Rochester
- 2023, University Research Award, University of Rochester
- 2023, ISCA Hall of Fame
- 2022, MICRO Hall of Fame
- 2020, NSF CAREER Award
- 2019, University Research Award, University of Rochester
- 2018, ACM SIGARCH – IEEE-CS TCCA Outstanding Dissertation Award, Honorable Mention
- 2017, Google Faculty Research Award
- 2016, Google Ph.D. Fellowship
- 2011, Microelectronics and Computer Development Fellowship, UT Austin

## PUBLICATION AWARDS

- 2025, ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), Best Paper Award  
“MetaSapiens: Real-Time Neural Rendering with Efficiency-Aware Pruning and Accelerated Foveated Rendering”
- 2022, IEEE Micro Top Picks of Computer Architecture, Honorable Mention  
“ANT: Exploiting Adaptive Numerical Data Type for Low-bit DNN Quantization”  
Originally published at MICRO 2022

- **2022, IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR), Best Paper Honorable Mention**  
*"Real-Time Gaze Tracking with Event-Driven Eye Segmentation"*
- **2022, IEEE International Symposium on High-Performance Computer Architecture (HPCA), Best Paper Nominee**  
*"S2TA: Exploiting Structured Sparsity for Energy-Efficient Mobile CNN Acceleration"*
- **2022, Kostas Pantazos Memorial Award for Outstanding Paper in Visualization and Data Analysis, Society for Imaging Science and Technology**  
*"Digital Reconstruction of Elmina Castle for Mobile Virtual Reality via Point-based Detail Transfer"*
- **2020, International Conference on Parallel Architectures and Compilation Techniques (PACT), Best Paper Nominee**  
*"Low-Latency Proactive Continuous Vision"*
- **2019, IEEE Micro Top Picks of Computer Architecture**  
*"Energy-Efficient Video Processing for Virtual Reality"*  
Originally published at ISCA 2019
- **2019, IEEE/ACM International Symposium on Microarchitecture (MICRO), Best Paper Nominee**  
*"Tigris: Architecture and Algorithms for 3D Perception in Point Clouds"*
- **2018, IEEE Micro Top Picks of Computer Architecture, Honorable Mention**  
*"Euphrates: Algorithm-SoC Co-Design for Low-Power Mobile Continuous Vision"*  
Originally published at ISCA 2018
- **2015, IEEE Micro Top Picks of Computer Architecture, Honorable Mention**  
*"Microarchitectural Implications of Event-driven Server-side Web Applications"*  
Originally published at MICRO 2015
- **2014, Best of Computer Architecture Letters Award**  
*"Exploiting Webpage Characteristics for Energy-Efficient Mobile Web Browsing"*
- **2010, Outstanding Undergraduate Thesis Award, Beihang University**  
*"Distributed Time, Conservative Parallel Logic Simulation on GPUs"*

## MENTORING AWARDS

- **2025, CRA Outstanding Undergraduate Researcher, Honorable Mention**  
*Junjie Zhao (Advisee)*
- **2020, CRA Outstanding Undergraduate Researcher, Honorable Mention**  
**2019, ACM Student Research Competition (ASPLOS 2019), Gold Medalist**  
*Sam Triest (Advisee)*
- **2019, CRA Outstanding Undergraduate Researcher, Honorable Mention**  
**2019, A. Richard Newton Young Student Fellow (DAC 2019)**  
*Qiuyue Sun (Advisee)*
- **2020, CRA Outstanding Undergraduate Researcher, Honorable Mention**  
*Sifan Ye (Advisee)*

## FUNDING

I have received a total funding of \$6.1 million, and my share is roughly \$3.6 million.

### Internal Funding

- University of Rochester, GIDS Seed Funding, \$30,000, 2025; Co-PI.
- University of Rochester, GIDS Seed Funding, \$30,000, 2024; Co-PI.

- University of Rochester, University Research Award, \$75,000, 2023; Co-PI.
- University of Rochester, GIDS Seed Funding, \$20,000, 2022; Lead PI.
- University of Rochester, University Research Award, \$75,000, 2019; Lead PI.

#### **External Industry Funding**

- Meta Research Grant, \$880,967 (my share: \$880,967), 2024 - 2028; Single PI.
- Meta Research Grant, \$577,254 (my share: \$391,760), 2021 - 2023; Lead PI.
- SRC AI Hardware (2021-AH-3023), \$210,000, 2021 - 2023; Single PI.
- Arm Research Gift Funding, \$85,000, 2020; Single PI.
- Google Gift Funding, \$50,000, 2018; Single PI.

#### **External Federal Funding**

- NSF ECCS #2419721, \$424,750 (my share: \$210,000), 2024 - 2027; Co-PI.
- NSF ENG EEC FuSe #2328856, \$1,999,480 (my share: \$499,480), 2024 - 2026; Co-PI.
- NSF CISE CNS #2225860, \$599,988 (my share: \$299,920), 2022 - 2025; Lead PI.
- NSF CISE CCF #2126642, \$499,988 (my share: \$285,042), 2021 - 2024; Lead PI.
- NEH HAA-280830-21, \$99,874 (my share: \$20,000), 2021 - 2023; Co-PI.
- NSF CISE CCF #2044963 (CAREER Award), \$515,680, 2021 - 2026; Single PI.

### **PROFESSIONAL POSITIONS**

#### **University of Rochester**

Associate Professor

- Department of Computer Science July 2024 - now
- Department of Brain and Cognitive Sciences July 2024 - now

Assistant Professor

- Department of Computer Science Jan. 2018 - July 2024
- Department of Brain and Cognitive Sciences Sept. 2023 - July 2024

Affiliated Faculty

- Goergen Institute for Data Science and AI June 2018 - now
- Center for Visual Science Sept. 2023 - now

#### **Reality Labs Research, Meta**

Visiting Researcher July 2025 - Feb. 2026

#### **Arm Research**

Visiting Researcher July 2017 - Dec. 2017

#### **Harvard University**

Research Fellow Aug. 2016 - May 2017

#### **The University of Texas at Austin**

Graduate Researcher Aug. 2010 - May 2017  
Teaching Assistant Fall 2010, Spring 2011, Spring 2014

#### **Google Inc.**

SWE Intern (Lead of [Flow API](#) as a part of the [Catapult](#) project) Summer 2015

#### **AMD Research Lab**

Research Intern Summer 2012, Summer 2013

STMicroelectronics

Co-op Engineer

Summer 2011

Tsinghua University

Undergraduate Researcher

June 2009 - May 2010

## PUBLICATIONS

### Journal Articles

- Shuang Wu, Bo Yu, Shaoshan Liu, [Yuhao Zhu](#)  
[Autonomy 2.0: The Quest for Economies of Scale](#)  
*Communications of the ACM*, 68.4 (2025): 28-32
- Zishen Wan, Yiming Gan, Bo Yu, Shaoshan Liu, Arijit Raychowdhury, [Yuhao Zhu](#)  
[The Vulnerability-Adaptive Protection Paradigm Toward Reliable Autonomous Machines](#)  
*Communications of the ACM*, 67.9 (2024): 66-77
- Yiming Gan, Jingwen Leng, Bo Yu, [Yuhao Zhu](#)  
[KINDRED: Heterogeneous Split-Lock Architecture for Safe Autonomous Machines](#)  
*ACM Transactions on Architecture and Code Optimization*, March 2025, 22(1):40
- Carlos Mauricio Villegas Burgos, Yu Feng, Pei Xiong, [Yuhao Zhu](#), Nick Vamivakas  
[Co-Designed Pre-capture Privacy Optics for Computer Vision](#)  
*Optical Express*, 33.12 (2025): 26590-26606
- Yu Feng, Weikai Lin, Zihan Liu, Jingwen Leng, Minyi Guo, Han Zhao, Xiaofeng Hou, Jieru Zhao, [Yuhao Zhu](#)  
[Potamoi: Accelerating Neural Rendering via a Unified Streaming Architecture](#)  
*ACM Transactions on Architecture and Code Optimization*, November 2024, 21(4):80
- Carlos Mauricio Villegas Burgos, Pei Xiong, Liangyu Qiu, [Yuhao Zhu](#), Nick Vamivakas  
[Co-designed Metaoptoelectronic Deep Learning](#)  
*Optical Express*, 31.4 (2023): 6453-6463
- Qiang Liu, Yuhui Hao, Weizhuang Liu, Bo Yu, Yiming Gan, Jie Tang, Shao-Shan Liu, [Yuhao Zhu](#)  
[An Energy Efficient and Runtime Reconfigurable Accelerator for Robotic Localization](#)  
*IEEE Transactions on Computers*, 2022, 72(7), 1943-1957
- Shaoshan Liu, Bo Yu, Jie Tang, [Yuhao Zhu](#), Xue Liu  
[Communication Challenges in Infrastructure-Vehicle Cooperative Autonomous Driving: A Field Deployment Perspective](#)  
*IEEE Wireless Communications*, 2022 May, 5;29(4):126-31
- Carlos Mauricio Villegas Burgos, Tianqi Yang, [Yuhao Zhu](#), Nick Vamivakas  
[A Design Framework for Metasurface Optics-based Convolutional Neural Networks](#)  
*Applied Optics*, 2021, (60)15:4356-4365
- Zishen Wan, Bo Yu, Thomas Yuang Li, Jie Tang, [Yuhao Zhu](#), Yu Wang, Arijit Raychowdhury, Shaoshan Liu  
[A Survey of FPGA-Based Robotic Computing](#)  
*IEEE Circuits and Systems Magazine*, 2021, 21(2):48-74

- Yue Leng, Chi-chun Chen, Qiuyue Sun, Jian Huang, [Yuhao Zhu](#)  
[Energy-Efficient Video Processing for Virtual Reality](#)  
*IEEE Micro Special Issue on Top Picks from the 2019 Computer Architecture Conferences*, May/June 2020, 40(3):30-36
- Yiming Gan, Yuxian Qiu, Jingwen Leng, [Yuhao Zhu](#)  
[SVSoC: Speculative Vision Systems-on-a-Chip](#)  
*Computer Architecture Letters (CAL)*, March 2019, 18(1):47-50
- [Yuhao Zhu](#), Vijay Janapa Reddi  
[Optimizing General-Purpose CPUs for Energy-Efficient Mobile Web Computing](#)  
*ACM Transactions on Computer Systems (TOCS)*, March 2017, 35(1):1
- [Yuhao Zhu](#), Vijay Janapa Reddi, Robert Adolf, Saketh Rama, Brandon Reagen, Gu-Yeon Wei, David M. Brooks  
[Cognitive Computing Safety: The New Horizon for Reliability / The Design and Evolution of Deep Learning Workloads](#)  
*IEEE Micro Special Issue on Cognitive Architectures*, Jan/Feb 2017, 37(1):15-21
- Peter Bailis, Jean Yang, Vijay Janapa Reddi, [Yuhao Zhu](#)  
[Research for Practice: Web Security and Mobile Web Computing \(Website\)](#)  
*Communications of the ACM*, Jan 2017, 60(1):50-53  
Also appears in *ACM Queue*, July/Aug 2016, 14(4):80-95.
- [Yuhao Zhu](#), Matthew Halpern, Vijay Janapa Reddi  
[The Role of the CPU in Energy-Efficient Mobile Web Browsing](#)  
*IEEE Micro Special Issue on Mobile Systems*, Jan/Feb 2015, 35(1):26-33
- [Yuhao Zhu](#), Aditya Srikanth, Jingwen Leng, Vijay Janapa Reddi  
[Exploiting Webpage Characteristics for Energy-Efficient Mobile Web Browsing](#)  
*Computer Architecture Letters (CAL)*, Oct 2012, 13(1):33-36  
**Awarded Best of Computer Architecture Letter in 2014**
- [Yuhao Zhu](#), Bo Wang, Yangdong Deng  
[Massively Parallel Logic Simulation with GPUs](#)  
*ACM Transactions on Design Automation of Electronic Systems (TODAES)*, June 2011, 16(3):29

### Conference Papers (Co)-Led By My Group

- Sheng Zhao, Weikai Lin, [Yuhao Zhu](#)  
[Multidimensional Observer Model and Perceptual Dimensions of Human Image Quality Assessment](#)  
*NeurIPS 2026*
- Sheng Zhao, Weikai Lin, [Yuhao Zhu](#)  
[GazeFlow: From Human Gaze Behavior to Generative Egocentric Gaze Prediction](#)  
*NeurIPS 2026*
- Ethan Chen, Xi Peng, Alexandre Chapiro, Dario Seyb, Michael Grundmann, Carll Marshall, Sushant Kondguli, Praneeth Chakravarthula, [Yuhao Zhu](#)  
[HoOPOE: A Holography-Optimized Perceptual Objective function](#)  
*SIGGRAPH Asia 2026*
- Weikai Lin, Junjie Zhao, Sushant Kondguli, Carll Marshall, [Yuhao Zhu](#)  
[ControlGS: Conditioning Neural Gaussians for Downstream-Processing-Aware](#)

## XR Rendering

*SIGGRAPH Asia 2026*

- Han Yan, Yifei Zou, Weikai Lin, Yuhao Zhu, Xuan Zhang  
[Invited Paper: A Quantitative Approach to Exploring Novel Image Sensor Architecture Towards Autonomous Edge Machine Vision](#)  
*DAC 2026*
- Abhishek Tyagi, Saurabh Hukerikar, Nirmal Saxena, Yanxiang Huang, Philip Shirvani, Chung-Hsuan Tung, Yuhao Zhu  
[LLM-PRISM: Characterizing Silent Data Corruption from Permanent GPU Faults in LLM Training](#)  
*SC 2026*
- Weikai Lin, Sheng Zhao, Ian Ross, Carl Marshall, Sushant Kondguli, Yuhao Zhu  
[LowPowAR: Power-Constrained Tone Mapping for Augmented Reality](#)  
*ISMAR (TVCG) 2026*
- Weikai Lin, Haoxiang Li, Yuhao Zhu  
[ControlHair: Synergizing Physics Simulator and Video Diffusion for Controllable Dynamic Hair Rendering](#)  
*ECCV 2026*
- Ethan Chen, Sushant Kondguli, Carl Marshall, Yuhao Zhu  
[Modeling and Exploiting the Time Course of Chromatic Adaptation for Display Power Optimizations in Virtual Reality](#)  
*SIGGRAPH Asia (TOG) 2025*
- Weikai Lin, Sushant Kondguli, Carl Marshall, Yuhao Zhu  
[PowerGS: Display-Rendering Power Co-Optimization for Neural Rendering in Power-Constrained XR Systems](#)  
*SIGGRAPH Asia 2025*
- Zhengyang Jin, Yuhao Zhu, T. Florian Jaeger  
[Latent Speech Representations Learned Through Self-supervised Learning Predict Listeners' Generalization of Adaptation across Talkers](#)  
*CogSci 2025*
- Abhishek Tyagi, Arjun Iyer, William H Renninger, Christopher Kanan, Yuhao Zhu  
[Dynamic Sparse Training of Diagonally Sparse Networks](#)  
*ICML 2025*
- Weikai Lin, Tianrui Ma, Adith Bloor, Yu Feng, Ruofan Xing, Xuan Zhang, Yuhao Zhu  
[SnapPix: Efficient-Coding-Inspired In-Sensor Compression for Edge Vision](#)  
*DAC 2025*
- Weikai Lin, Yu Feng, Yuhao Zhu  
[MetaSapiens: Real-Time Neural Rendering with Efficiency-Aware Pruning and Accelerated Foveated Rendering](#)  
*ASPLOS 2025*  
**Best Paper Award**
- Ethan Chen, Jiwon Chang, Yuhao Zhu  
[CoolerSpace: A Language for Physically Correct and Computationally Efficient Color Programming](#)  
*OOPSLA 2024*

- Yuhao Zhu  
Invited Paper: Imaging, Computing, and Human Perception: Three Agents to Usher in the Autonomous Machine Computing Era  
ICCAD 2024
- Yuhao Zhu, Ethan Chen, Colin Hascup, Yukang Yan, Gaurav Sharma  
Computational Trichromacy Reconstruction: Empowering the Color-Vision Deficient to Recognize Colors Using Augmented Reality  
UIST 2024
- Yu Feng, Tianrui Ma, Yuhao Zhu, Xuan Zhang  
BlissCam: Boosting Eye Tracking Efficiency with Learned In-Sensor Sparse Sampling  
ISCA 2024
- Yu Feng, Zihan Liu, Jingwen Leng, Minyi Guo, Yuhao Zhu  
Cicero: Addressing Algorithmic and Architectural Bottlenecks in Neural Rendering by Radiance Warping and Memory Optimizations  
ISCA 2024
- Nisarg Ujjainkar, Ethan Shahan, Kenneth Chen, Budmonde Duinkharjav, Qi Sun, Yuhao Zhu  
Exploiting Human Color Discrimination for Memory- and Energy-Efficient Image Encoding in Virtual Reality  
ASPLOS 2024
- Abhishek Tyagi, Reiley Jeyapaul, Chuteng Zhou, Paul Whatmough, Yuhao Zhu  
Characterizing Soft-Error Resiliency in Arm's Ethos-U55 Embedded Machine Learning Accelerator  
ISPASS 2024
- Yu Feng, Tianrui Ma, Adith Boloor, Yuhao Zhu, Xuan Zhang  
Invited Paper: Learned In-Sensor Visual Computing: From Compression to Eventification  
ICCAD 2023
- Kenneth Chen, Budmonde Duinkharjav, Nisarg Ujjainkar, Ethan Shahan, Abhishek Tyagi, Jiayi He, Yuhao Zhu, Qi Sun  
Imperceptible Color Modulation for Power Saving in VR/AR  
SIGGRAPH Emerging Technologies 2023
- Tianrui Ma, Yu Feng, Xuan Zhang, Yuhao Zhu  
CamJ: Enabling System-Level Energy Modeling and Architectural Exploration for In-Sensor Visual Computing  
ISCA 2023
- Nisarg Ujjainkar, Jingwen Leng, Yuhao Zhu  
ImaGen: A General Framework for Generating Memory- and Power-Efficient Image Processing Accelerators  
ISCA 2023
- Yuhui Hao, Yiming Gan, Bo Yu, Qiang Liu, Shaoshan Liu, Yuhao Zhu  
BLITZCRANK: Factor Graph Accelerator for Motion Planning  
DAC 2023
- Yuhao Zhu  
Teaching Color Science to EECS Students Using Interactive Tutorials: Tools and

## Lessons

*IS&T EI (VDA) 2023*

- Elias Neuman-Donihue, Michael Jarvis, Yuhao Zhu  
[FastPoints: A State-of-the-Art Point Cloud Renderer for Unity](#)  
*IS&T EI (VDA) 2023*
- Abhishek Tyagi, Yiming Gan, Shaoshan Liu, Bo Yu, Paul Whatmough, Yuhao Zhu  
[Thales: Formulating and Estimating Architectural Vulnerability Factors for DNN Accelerators](#)  
*HPCA 2023*
- Yu Feng, Patrick Hansen, Paul N. Whatmough, Guoyu Lu, Yuhao Zhu  
[Fast and Accurate: Video Enhancement Using Sparse Depth](#)  
*WACV 2023*
- Budmonde Duinkharjav, Kenny Chen, Abhishek Tyagi, Jiayi He, Yuhao Zhu, Qi Sun  
[Color-Perception-Guided Display Power Reduction for Virtual Reality](#)  
*SIGGRAPH Asia (TOG) 2022*
- Yu Feng, Gunnar Hammonds, Yiming Gan, Yuhao Zhu  
[Crescent: Taming Memory Irregularities for Accelerating Deep Point Cloud Analytics](#)  
*ISCA 2022*
- Gregory Heyworth, Keith T. Knox, Kenneth Boydston, Yuhao Zhu  
[Multispectral Scheimpflug: Imaging Degraded Books That Open Less Than 30 Degrees](#)  
*IS&T Archiving 2022*
- Yiming Gan, Paul Whatmough, Bo Yu, Shaoshan Liu, Yuhao Zhu  
[BRAUM: Analyzing and Protecting Autonomous Machine Software Stack](#)  
*ISSRE 2022*
- Yu Feng, Nathan Goulding-Hotta, Asif Khan, Hans Reyserhove, Yuhao Zhu  
[Real-Time Gaze Tracking with Event-Driven Eye Segmentation](#)  
*IEEE VR 2022*  
**Best Paper Nominee; Invited Presentation at IEEE VIS 2022**
- Yuhao Zhu  
[RTNN: Accelerating Neighbor Search Using Hardware Ray Tracing](#)  
*PPoPP 2022*
- Sifan Ye, Ting Wu, Michael Jarvis, Yuhao Zhu  
[Digital Reconstruction of Elmina Castle for Mobile Virtual Reality via Point-based Detail Transfer](#)  
*IS&T EI (VDA) 2022*  
**Kostas Pantazos Memorial Award for Outstanding Paper in Visualization and Data Analysis**
- Weizhuang Liu, Bo Yu, Yiming Gan, Qiang Liu, Jie Tang, Shaoshan Liu, Yuhao Zhu  
[Archytas: A Framework for Synthesizing and Dynamically Optimizing Accelerators for Robotic Localization](#)  
*MICRO 2021*
- Joshua Romphf, Elias Neuman-Donihue, Gregory Heyworth, Yuhao Zhu

[Resurrect3D: An Open and Customizable Platform for Visualizing and Analyzing Cultural Heritage Artifacts](#)

*Web3D 2021*

- Yiming Gan, Bo Yu, Boyuan Tian, Leimeng Xu, Wei Hu, Shaoshan Liu, Qiang Liu, Yanjun Zhang, Jie Tang, [Yuhao Zhu](#)  
[Eudoxus: Characterizing and Accelerating Localization in Autonomous Machines](#)  
*HPCA 2021*
- Yu Feng, Boyuan Tian, Tiancheng Xu, Paul Whatmough, [Yuhao Zhu](#)  
[Mesorasi: Architecture Support for Point Cloud Analytics via Delayed-Aggregation](#)  
*MICRO 2020*
- Bo Yu, Wei Hu, Leimeng Xu, Jie Tang, Shaoshan Liu, [Yuhao Zhu](#)  
[Building the Computing System for Autonomous Micromobility Vehicles: Design Constraints and Architectural Optimizations](#)  
*MICRO 2020*
- Yiming Gan, Yuxian Qiu, Jingwen Leng, Minyi Guo, [Yuhao Zhu](#)  
[Ptolemy: Architecture Support for Robust Deep Learning](#)  
*MICRO 2020*
- Yu Feng, Shaoshan Liu, [Yuhao Zhu](#)  
[Real-Time Spatio-Temporal LiDAR Point Cloud Compression](#)  
*IROS 2020*
- Haichuan Yang, Shupeng Gui, [Yuhao Zhu](#), Ji Liu  
[Automatic Neural Network Compression by Sparsity-Quantization Joint Learning: A Constrained Optimization-based Approach](#)  
*CVPR 2020*
- Yiming Gan, Yuxian Qiu, Lele Chen, Jingwen Leng, [Yuhao Zhu](#)  
[Low-Latency Proactive Continuous Vision](#)  
*PACT 2020*  
**Best Paper Nominee**
- Qiuyue Sun, Amir Taherin, Yawo Siatitse, [Yuhao Zhu](#)  
[Energy-Efficient 360-Degree Video Rendering on FPGA via Algorithm-Architecture Co-Design](#)  
*FPGA 2020*
- Yu Feng, Paul Whatmough, [Yuhao Zhu](#)  
[ASV: Accelerated Stereo Vision System](#)  
*MICRO 2019*
- Tiancheng Xu, Boyuan Tian, [Yuhao Zhu](#)  
[Tigris: Architecture and Algorithms for 3D Perception in Point Clouds](#)  
*MICRO 2019*  
**Best Paper Nominee**
- Yu Feng, [Yuhao Zhu](#)  
[PES: Proactive Event Scheduling for Energy-Efficient Mobile Web Computing](#)  
*ISCA 2019*
- Yue Leng, Chi-chun Chen, Qiuyue Sun, Jian Huang, [Yuhao Zhu](#)  
[Energy-Efficient Video Processing for Virtual Reality](#)  
*ISCA 2019*  
**IEEE Micro Top Picks of Computer Architecture in 2019**

- Haichuan Yang, [Yuhao Zhu](#), Ji Liu  
[ECC: Energy-Constrained Deep Neural Network Compression via a Bilinear Regression Model](#)  
*CVPR 2019*
- Haichuan Yang, [Yuhao Zhu](#), Ji Liu  
[Energy-Constrained Compression for Deep Neural Networks via Weighted Sparse Projection and Layer Input Masking](#)  
*ICLR 2019*
- Yue Leng, Chi-chun Chen, Qiuyue Sun, Jian Huang, [Yuhao Zhu](#)  
[Semantic-Aware Virtual Reality Video Streaming](#)  
*APSys 2018*
- [Yuhao Zhu](#), Anand Samajdar, Matthew Mattina, Paul Whatmough  
[Euphrates: Algorithm-SoC Co-Design for Low-Power Mobile Continuous Vision](#)  
*ISCA 2018*  
**IEEE Micro Top Picks of Computer Architecture (Honorable Mention) in 2018**
- [Yuhao Zhu](#), Matthew Mattina, Paul Whatmough  
[Mobile Machine Learning Hardware at ARM: A Systems-on-Chip \(SoC\) Perspective](#)  
*SysML 2018*

### Conference Papers My Group Contribute To

- Ruofan Xing, Arman Akbari, Weikai Lin, Adith Bloor, Alexander Montes McNeil, Michael Moebius, Yongmin Liu, [Yuhao Zhu](#), Xuan Zhang  
[HoloCode: Hybrid Optical-Electronic Edge Encoding for Privacy-Preserving Cloud Training](#)  
*DAC 2026*
- Yu Feng, Weikai Lin, Yuge Cheng, Zihan Liu, Jingwen Leng, Minyi Guo, Chen Chen, Shixuan Sun, [Yuhao Zhu](#)  
[Lumina: Real-Time Neural Rendering by Exploiting Computational Redundancy](#)  
*ISCA 2025*
- Ryan Hou, Thomas Twomey, Vasileios Milionis, Evangelos Dikopoulos, Tianrui Ma, [Yuhao Zhu](#), Georgios Tzimpragos  
[SEAL: A Single-Event Architecture for In-Sensor Visual Localization](#)  
*ISCA 2025*
- Yu Feng, Zheng Liu, Weikai Lin, Zihan Liu, Jingwen Leng, Minyi Guo, Zhezhi He, Jieru Zhao, [Yuhao Zhu](#)  
[StreamGrid: Streaming Point Cloud Analytics via Compulsory Splitting and Deterministic Termination](#)  
*ASPLOS 2025*
- Zihan Liu, Xinhao Luo, Junxian Guo, Wentao Ni, Yangjie Zhou, Yue Guan, Cong Guo, Weihao Cui, Yu Feng, Minyi Guo, [Yuhao Zhu](#), Minjia Zhang, Jingwen Leng, Chen Jin  
[VQ-LLM: High-performance Code Generation for Vector Quantization Augmented LLM Inference](#)  
*HPCA 2025*
- Adith Bloor, Weikai Lin, Tianrui Ma, Yu Feng, [Yuhao Zhu](#), Xuan Zhang

[PrivateEye: In-Sensor Privacy Preservation Through Optical Feature Separation](#)  
WACV 2025

- Shaoshan Liu, [Yuhao Zhu](#), Bo Yu, Jean-Luc Gaudiot, Guang R. Gao  
[Invited Paper: Dataflow Accelerator Architecture for Autonomous Machine Computing](#)  
ICCAD 2024
- Yue Guan, Yuxian Qiu, Jingwen Leng, Fan Yang, Shou Yu, Yunxin Liu, Yu Feng, [Yuhao Zhu](#), Lidong Zhou, Yun Liang, Chen Zhang, Chao Li, Minyi Guo  
[Amanda: Unified Instrumentation Framework for Deep Neural Networks](#)  
ASPLOS 2024
- Zihan Liu, Wentao Ni, Jingwen Leng, Yu Feng, Cong Guo, Quan Chen, Chao Li, Minyi Guo, [Yuhao Zhu](#)  
[Juno: Optimizing High-Dimensional Approximate Nearest Neighbour Search with Sparsity-Aware Algorithm and Ray-Tracing Core Mapping](#)  
ASPLOS 2024
- Cong Guo, Jiaming Tang, Weiming Hu, Jingwen Leng, Chen Zhang, Fan Yang, Yunxin Liu, Minyi Guo, [Yuhao Zhu](#)  
[OliVe: Accelerating Large Language Models via Hardware-friendly Outlier-Victim Pair Quantization](#)  
ISCA 2023
- Cong Guo, Chen Zhang, Jingwen Leng, Zihan Liu, Fan Yang, Yunxin Liu, Minyi Guo, [Yuhao Zhu](#)  
[ANT: Exploiting Adaptive Numerical Data Type for Low-bit Deep Neural Network Quantization](#)  
MICRO 2022  
**IEEE Micro Top Picks of Computer Architecture (Honorable Mention) in 2022**
- Zhi-Gang Liu, Paul Whatmough, [Yuhao Zhu](#), Matt Mattina  
[S2TA: Exploiting Structured Sparsity for Energy-Efficient Mobile CNN Acceleration](#)  
HPCA 2022  
**Best Paper Nominee**
- Yue Guan, Zhengyi Li, Zhouhan Lin, [Yuhao Zhu](#), Jingwen Leng, Minyi Guo  
[Block-Skim: Efficient Question Answering for Transformer](#)  
AAAI 2022
- Cong Guo, Yuxian Qiu, Jingwen Leng, Xiaotian Gao, Chen Zhang, Yunxin Liu, Fan Yang, [Yuhao Zhu](#), Minyi Guo  
[SQuant: On-the-Fly Data-Free Quantization via Diagonal Hessian Approximation](#)  
ICLR 2022
- Yangjie Zhou, Mengtian Yang, Cong Guo, Jingwen Leng, Yun Liang, Quan Chen, Minyi Guo, [Yuhao Zhu](#)  
[Characterizing and Demystifying the Implicit Convolution Algorithm on Commercial Matrix-Multiplication Accelerators](#)  
IISWC 2021
- Shaoshan Liu, Bo Yu, Yahui Liu, Kunai Zhang, Yisong Qiao, Thomas Yuang Li, Jie Tang, [Yuhao Zhu](#)  
[Brief Industry Paper: The Matter of Time — A General and Efficient System for](#)

Precise Sensor Synchronization in Robotic Computing  
RTAS 2021

- Cong Guo, Bo Yang Hsueh, Jingwen Leng, Yuxian Qiu, Yue Guan, Zehuan Wang, Xiaoying Jia, Xipeng Li, Minyi Guo, Yuhao Zhu  
[Accelerating Sparse DNN Models Without Hardware-Support via Tile-wise Sparsity](#)  
SC 2020
- Cong Guo, Yangjie Zhou, Jingwen Leng, Yuhao Zhu, Zidong Du, Quan Chen, Chao Li, Minyi Guo, Bin Yao  
[Balancing Efficiency and Flexibility for DNN Acceleration via Temporal GPU-Systolic Array Integration](#)  
DAC 2020
- Anand Samajdar, J. Joseph, Yuhao Zhu, Paul Whatmough, Matt Mattina, Tushar Krishna  
[A Systematic Methodology for Characterizing Scalability of DNN Accelerators](#)  
ISPASS 2020
- Yuxian Qiu, Jingwen Leng, Cong Guo, Quan Chen, Chao Li, Minyi Guo, Yuhao Zhu  
[Adversarial Defense Through Network Profiling Based Path Extraction](#)  
CVPR 2019
- Yu Wang, Yuhao Zhu, Glenn Ko, Brandon Reagen, Gu-Yeon Wei, David Brooks  
[Demystifying Bayesian Inference Workloads](#)  
ISPASS 2019
- Yuwei Hu, Jidong Zhai, Dinghua Li, Yifan Gong, Yuhao Zhu, Wei Liu, Lei Su, Jiangming Jin  
[BitFlow: Exploiting Vector Parallelism for Binary Neural Networks on CPU](#)  
IPDPS 2018

**Pre-2018 Conference Papers**

- Wenzhi Cui, Daniel Richins, Yuhao Zhu, Vijay Janapa Reddi  
[Tail Latency in Node.js: Energy Efficient Turbo Boosting for Long Latency Requests in Event-Driven Web Services](#)  
VEE 2019
- Yuhao Zhu, Vijay Janapa Reddi  
[GreenWeb: Language Extensions for Energy-Efficient Mobile Web Computing](#)  
PLDI 2016
- Matthew Halpern, Yuhao Zhu, Vijay Janapa Reddi  
[Mobile CPU's Rise to Power: Quantifying the Impact of Generational Mobile CPU Design Trends on Performance, Energy, and User Satisfaction](#)  
HPCA 2016
- Yuhao Zhu, Daniel Richins, Matthew Halpern, Vijay Janapa Reddi  
[Microarchitectural Implications of Event-driven Server-side Web Applications](#)  
MICRO 2015  
**IEEE Micro Top Picks of Computer Architecture (Honorable Mention) in 2015**
- Yuhao Zhu, Matthew Halpern, Vijay Janapa Reddi  
[Event-based Scheduling for Energy-Efficient QoS \(eQoS\) in Mobile Web Applications](#)

#### HPCA 2015

- Matthew Halpern, [Yuhao Zhu](#), Ramesh Peri, Vijay Janapa Reddi  
[Mosaic: Cross-Platform User-Interaction Record and Replay Tool for the Fragmented Android Ecosystem](#)  
ISPASS 2015
- [Yuhao Zhu](#), Vijay Janapa Reddi  
[WebCore: Architectural Support for Mobile Web Browsing](#)  
ISCA 2014
- Chen Zhou, Xiaofei Wang, Weichao Xu, [Yuhao Zhu](#), Vijay Janapa Reddi, Chris Kim  
[Estimation of Instantaneous Frequency Fluctuation in a Fast DVFS Environment Using an Empirical BTI Stress-Relaxation Model](#)  
IRPS 2014
- [Yuhao Zhu](#), Vijay Janapa Reddi  
[High-Performance and Energy-Efficient Mobile Web Browsing on Big/Little Systems](#)  
HPCA 2013
- [Yuhao Zhu](#), Yangdong Deng, Yubei Chen  
[Hermes: An Integrated CPU/GPU Microarchitecture for IP Routing](#)  
DAC 2011
- Bo Wang, [Yuhao Zhu](#), Yangdong Deng  
[Distributed Time, Conservative Parallel Logic Simulation on GPUs](#)  
DAC 2010

#### Patents

- [Yuhao Zhu](#), Qi Sun, Nisarg Ujjainkar, Ethan Shahan, Kenneth Chen, Budmonde Duinkharjav  
[Human Color Discrimination for Memory-Efficient Encoding in Extended Reality](#)  
US Patent App. 18/884,550
- [Yuhao Zhu](#), Paul Whatmough  
[Computer Vision Processing](#)  
US Patent App. 16/127,007
- [Yuhao Zhu](#), Paul Whatmough  
[Region of Interest Determination in Video](#)  
US Patent App. 15/875,464

#### Books and Book Chapters

- [Yuhao Zhu](#)  
[Visual Computing For Architects: Imaging, Rendering, and Human Vision](#)  
Part of the *Synthesis Lectures on Computer Architecture* series  
Springer Nature, 2026
- Yangdong Deng, [Yuhao Zhu](#), Bo Wang  
[Asynchronous Parallel Logic Simulation on Modern Graphics Processors](#)  
In Book *GPU Solutions to Multi-scale Problems in Science and Engineering*  
Springer Nature, 2013

- Yangdong Deng, Xiaomeng Jiao, Shuai Mu, Kang Kang, [Yuhao Zhu](#)  
[NPGPU: Network Processing on Graphics Processing Units](#)  
In Book *Theoretical and Mathematical Foundations of Computer Science*  
Spring Nature, 2011

## Invited Talks

- **Perceptually-Driven Power-Efficient Rendering in Immersive Computing**  
*Universidad de Zaragoza, June 2026, Zaragoza, Spain*
- **Modeling Chromatic Adaptation and Image Quality Judgment in Humans**  
*University of Cambridge, June 2026, Cambridge, UK*
- **Jointly Optimizing Computer Architecture with Human Vision**  
*Imperial College London, June 2026, London, UK*
- **Modeling Human Color Vision For Science and Engineering**  
*Meta, Feb. 2026, Redmond, WA*
- **Color Discrimination and Chromatic Adaptation: Measuring, Modeling, and Exploiting Them for Energy-Efficient XR Displays**  
*Optica Color Technical Group Webinar Series, Sept. 2025, Online*
- **The Kinetics of Chromatic Adaptation Under Dynamic Lighting**  
*Munsell Color Science Laboratory, RIT, June 2025, Rochester, NY*  
*Meta, July 2025, Sunnyvale, CA*  
*Wu Tsai Neurosciences Institute, Stanford University, Aug. 2025, Palo Alto, CA*  
*Department of Psychology, University of Pennsylvania, Jan. 2026, Philadelphia, PA*
- **Efficient-Coding-Inspired In-Sensor Compression for Visual Computing**  
*MADCAP Workshop co-located with FCCM 2025, May 2025, Fayetteville, AR*
- **Moving Towards Reliable Autonomous Machines: The Vulnerability-Adaptive Protection Paradigm**  
*RoboArch Workshop co-located with MICRO 2024, Nov 2024, Austin, TX*
- **Imaging, Computing, and Human Perception: Three Agents to Usher in the Autonomous Machine Computing Era**  
*ICCAD 2024, Special Session on The Dawn of Domain-Specific Hardware System for Autonomous Machines, Oct 2024, New Jersey*
- **Human Visual Perception Meets AR/VR**  
*AR/VR Symposium, University of Rochester, Oct 2024, Rochester, NY*
- **Perceptual Rendering Meets Neural Rendering and Architecture**  
*Meta, Oct 2024, Virtual*
- **Harnessing Imaging, Computing, and Biological Perception Symbiosis**  
*Cornell University, Oct 2023, Ithaca, NY*  
*The University of Texas at Austin, Oct 2023, Austin, TX*  
*University of Pennsylvania, Oct 2023, Philadelphia, PA*  
*Duke University, Oct 2023, Durham, NC*  
*University of California, Los Angeles, Oct 2023, Los Angeles, CA*  
*University of California, Irvine, Oct 2023, Irvine, CA*  
*University of California, Riverside, Oct 2023, Riverside, CA*  
*University of Cambridge, Sept 2023, Cambridge, UK*

- **Harnessing and Harvesting the Computer Science-Vision Science Symbiosis**  
*University of North Carolina, Nov 2024, Chapel Hill, NC*  
*School of Optometry and Vision Science, UC Berkeley, Jan 2024, Berkeley, CA*  
*Center for Perceptual Systems, The University of Texas at Austin, Oct 2023, Austin, TX*  
*University of California, Santa Barbara, Oct 2023, Santa Barbara, CA*  
*American University, Oct 2023, Washington, D.C.*  
*National Eye Institute, Oct 2023, Washington, D.C.*  
*Dept. of Brain and Cognitive Sciences, University of Rochester, Aug 2023, Rochester, NY*  
*Institute of Ophthalmology, University College London, Sept 2023, London, UK*  
*Color Impact 2023, June 2023, Rochester, NY*  
*Munsell Color Science Laboratory, RIT, March 2023, Rochester, NY*  
*Wu Tsai Neurosciences Institute, Stanford University, April 2023, Palo Alto, CA*  
*Smith Kettlewell Eye Research Institute, April 2023, San Francisco, CA*
- **What Do We Need to Achieve a Scalable Autonomy Economy?**  
*RoboArch Workshop co-located with MICRO 2023, Oct 2023, Toronto, Canada*
- **Dark Silicon: What It Is, How We Got Here, and How We Get Out of It**  
*Dept. of Physics and Astronomy, University of Rochester, Oct 2023, Rochester, NY*
- **Rethinking Imaging-Computing Interface**  
*ICCAD 2023, Special Session on In-Sensor AI Computing Towards Next Generation Autonomous Edge Intelligence, Nov 2023, Santa Clara, CA*  
*Omnivision, Nov 2023, Santa Clara, CA*  
*Stanford Center for Image Systems Engineering, April 2023, Palo Alto, CA*  
*University College London, March 2023, Virtual*
- **High Performance Embedded Imaging: An Optics, Sensing, and Computing Co-Designed Approach**  
*High Performance Computing for Imaging at EI 2023, Jan 2023, San Francisco, CA*
- **Color Perception-Guided Display Optimizations: Power and Beyond**  
*Meta, Sept 2022, Virtual*
- **Go Horizontal, Not Vertical: Addressing Visual Computing Challenges in Autonomous Machines**  
*RoboArch Workshop co-located with MICRO 2022, Oct 2022, Chicago, IL*
- **RTNN: Accelerating Neighbor Search Using Hardware Ray Tracing**  
[\*WDDSA Workshop\*](#) co-located with MICRO 2022, Oct 2022, Chicago, IL
- **Improving and Harnessing Software Resiliency in Autonomous Machines**  
*Plenary Panel on Reliability of Autonomous Machines, IEEE COMPSAC, June 2022, Virtual*
- **Visual Computing: A Horizontal Approach**  
*Efficient AI Seminar, Rutgers University, Oct 2022, Virtual*  
*Keynote Speech, IFIP NPC, Sept 2022, Virtual*  
*GlobalFoundries, May 2022, Virtual*  
*University of Massachusetts Amherst, May 2022, Virtual*
- **Intelligent Visual Computing**  
*Duke University, March 2021, Virtual*  
*University of Utah, Oct 2021, Virtual*

*HALO workshop co-located with ICCAD, Nov 2021, Virtual*

- [Architecture Support for Robust Deep Learning: Exploiting Software 1.0 Techniques to Defend Software 2.0](#)  
*AMD Research, Oct 2020, Virtual*
- **Rethinking Computer Systems Stack for Point Cloud Processing**  
*Arm Research, Sept 2019, Waltham, MA*
- **Getting Computer Systems Ready for Visual Computing in Ten Years**  
*Intel Labs, Sept 2019, Hillsboro, OR*  
*Harvard University, Sept 2019, Cambridge, MA*  
*Yale University, Oct 2019, New Haven, CT*  
*University of Michigan, Nov 2019, Ann Arbor, MI*  
*FastPath Workshop co-located with ISPASS, Aug 2020, Virtual*
- **The Next Quintillion Pixels and Beyond: Architecting Next-Generation Mobile Visual Computing Systems**  
*Arm Research, Sept 2019, Austin, TX*  
*UT Austin, Sept 2019, Austin, TX*  
*Rice University, Sept 2019, Houston, TX*
- **Resource-Guaranteed Deep Learning**  
*Arm Research, April 2019, Waltham, MA*
- **Energy-Efficient Mobile Web: Proactive and Reactive Perspectives**  
*Google, April 2018, Seattle, WA*
- **Algorithm-SoC Co-design for Energy-Efficient Mobile Continuous Vision**  
*Cornell University, Feb 2018, Ithaca, NY*  
*CogArch Workshop co-located with ASPLOS 2018, March 2018, Williamsburg, VA*  
*SRI International, April 2018, Princeton, NJ*  
*Rochester Institute of Technology, Feb 2019, Rochester, NY*
- **The Watt Wise Web**  
*Texas A&M University, Jan 2017, Teleseminar*  
*Boston Area Architecture Workshop (BARC), Jan 2017, Cambridge, MA*  
*ARM Research, Jan 2017, Austin, TX*  
*UT Austin School of Information, Feb 2017, Austin, TX*
- **WebCore: Architectural Support for Mobile Web Browsing**  
*Intel, July 2014, Austin, TX*
- **High-Performance and EnergyEfficient Mobile Web Browsing on Big/Little Systems**  
*UT Austin Programming Language Lunch Seminar, September 2012, Austin, TX*  
*AMD Research Lab, August 2012, Austin, TX*

## Conference Presentations

- **Efficient-Coding-Inspired In-sensor Compression**  
*Computational Imaging and Machine Learning for Scientific Imaging at EI 2026, March 2026, Burlingame, CA*
- **The Kinetics of Chromatic Adaptation Under Dynamic Lighting**  
*Human Vision and Electronic Imaging at EI 2026, March 2026, Burlingame, CA*

- Computational Trichromacy Reconstruction: Empowering the Color-Vision Deficient to Recognize Colors Using Augmented Reality  
*Human Vision and Electronic Imaging at EI 2025*, Feb. 2025, Burlingame, CA
- Color Perception-Driven Energy Optimization for Virtual Reality: Displays and Image Encoding  
*Human Vision and Electronic Imaging at EI 2024*, Jan 2024, Burlingame, CA
- Modeling and Reducing Energy Consumption of Computational Image Sensors: A Case Study on Gaze Tracking  
*Image Sensors and Systems at EI 2024*, Jan 2024, Burlingame, CA
- Teaching Color Sciences to EECS Students Using Interactive Tutorials  
*Visual Data Analytics at EI 2023*, Jan 2023, San Francisco, CA
- [RTNN: Accelerating Neighbor Search Using Hardware Ray Tracing](#)  
*PPoPP 2022*, April 2022, Virtual
- [Tail Latency in Node.js: Energy Efficient Turbo Boosting for Long Latency Requests in Event-Driven Web Services](#)  
*VEE 2019*, April 2019, Providence, RI
- [Algorithm-SoC Co-design for Energy-Efficient Mobile Continuous Vision](#)  
*ISCA 2018*, June 2018, Los Angeles, CA
- [GreenWeb: Language Extensions for Energy-Efficient Mobile Web Computing](#)  
*PLDI 2016*, June 2016, Santa Barbara, CA
- [Energy and Power Measurement on Mobile Devices](#)  
*MobiTools co-located with ISCA 2016*, June 2016, Seoul, Korea
- [The Human Processing Unit \(HPU\) as a New Approximate Computing Substrate](#)  
*WAX 2016 co-located with ASPLOS 2016*, April 2016, Atlanta, GA
- [Scalable End-to-end Quality Control in Approximate Computing](#)  
*WAX 2016 co-located with ASPLOS 2016*, April 2016, Atlanta, GA
- [Microarchitectural Implications of Event-driven Server-side Web Applications](#)  
*MICRO 2015*, December 2015, Waikiki, HI ([lightening version](#))
- [Exploiting Webpage Characteristics for Energy-Efficient Mobile Web Browsing](#)  
*Best of CAL 2014 presented at HPCA 2015*, February 2015, San Francisco, CA
- [Event-Based Scheduling for Energy-Efficient QoS \(eQoS\) in Mobile Web Applications](#)  
*HPCA 2015*, February 2015, San Francisco, CA
- [WebCore: Architectural Support for Mobile Web Browsing](#)  
*ISCA 2014*, June 2014, Minneapolis, MN ([lightening version](#))
- [High-Performance and EnergyEfficient Mobile Web Browsing on Big/Little Systems](#)  
*HPCA 2013*, February 2013, Shenzhen, China
- [Hermes: An Integrated CPU/GPU Microarchitecture for IP Routing](#)  
*DAC 2011*, June 2011, San Diego, CA

## Guest Lectures

- **The Deficiencies and Evolution of Color Vision**  
*NYU, Data Visualization, Spring 2026*  
*NYU, Data Visualization, Spring 2024*
- **Modeling Light-Matter Interactions for Computer Graphics**  
*University of Rochester, Problem Seminar (Introduction to Graduate Study), Fall 2025*  
*University of Rochester, Undergraduate Problem Seminar, Spring 2025*
- **Human Color Vision and Its Implications on Computer Systems**  
*University of Rochester, Undergraduate Problem Seminar, Spring 2024*
- **Digital Camera Imaging**  
*University of Rochester, Undergraduate Problem Seminar, Spring 2023*
- **Accelerating Neighbor Search Using Hardware Ray Tracing**  
*University of Rochester, Undergraduate Problem Seminar, Spring 2022*
- **Things I Wish I Knew When I Started Doing Research**  
*University of Rochester, Problem Seminar (Introduction to Graduate Study), Fall 2021*
- **Watt-Wise Web: Architecting a Responsive and Energy-Efficient Mobile Web**  
*University of Utah, Web Browser Internals, Fall 2020*
- **Introduction to Computer Architecture Research**  
*University of Rochester, Problem Seminar (Introduction to Graduate Study), Fall 2018*
- **Architectural Support for Mobile Web**  
*Harvard University, Mixed-Signal IC Design, Fall 2016*
- **Mobile CPU Design Trends**  
*UT Austin, Computer Architecture: User System Interplay, Spring 2016*
- **Compilation for Power-Efficiency**  
*UT Austin, Dynamic Compilation, Spring 2014*
- **Garbage Collection**  
*UT Austin, Dynamic Compilation, Spring 2014*
- **Model Checking**  
*UT Austin, Advanced Compiler, Fall 2010*

## SOFTWARE

See the [GitHub page](#) of the Horizon Lab.

## EXTERNAL SERVICE

### Program Committee

- [ISCA](#): 2025, 2024, 2023 (ERC), 2022 (ERC), 2020, 2019, 2017 (ERC)
- [HPCA](#): 2024 (LPC), 2022, 2019, 2018 (ERC)
- [MICRO](#): 2026 (LPC), 2022 (+SRC), 2021 (ERC), 2020 (ERC), 2019
- [ASPLOS](#): 2027 (Fall), 2026, 2025, 2024
- [SC](#): 2022
- [IPDPS](#): 2023
- [HPCI](#): 2022

- [IEEE ISMAR](#): 2026, 2025
- [IEEE VR](#): 2027, 2026, 2025, 2023
- [ECCV](#): 2026, 2022
- [ICCV](#): 2023, 2021
- [CVPR](#): 2022, 2021, 2020
- [ICLR](#): 2022, 2021, 2020
- [AAAI](#): 2023, 2022, 2021
- [NeurIPS](#): 2026, 2022, 2021, 2020
- [WACV](#): 2021
- [ACCV](#): 2020
- [ISLPED](#): 2021
- [IISWC](#): 2021, 2019
- [ISPASS](#): 2022, 2019
- IEEE Micro Top Picks: 2018
- [CGO-PPoPP Artifacts Evaluation](#): 2016, 2015
- [TinyToCS](#) Volume IV, III

#### Solicited Reviewer

- [VRST](#) (2026), [ACM TOG](#) (2025), [IEEE Access](#) (2025), [Nature Communication](#) (2025), [SIGGRAPH Asia](#) (2026), [SIGGRAPH](#) (2026, 2025), [PPoPP](#) (2025), [ACM TECS](#) (2025, 2024), [UIST](#) (2024), [IEEE TVCG](#) (2024, 2023), [JOSA A](#) (2023), [IROS](#) (2023), [IEEE TPAMI](#) (2023), [IEEE TCAD](#) (2023), [IEEE VR](#) (2021), [IEEE TCC](#) (2022), [IEEE ISMAR](#) (2023, 2022), [ACM TOCS](#) (2022, 2019), [ASPLOS](#) (2027 (Spring), 2021), [IEEE TNNLS](#) (2021, 2020), [IEEE TSUSC](#) (2020), [ACM TWEB](#) (2020), [IEEE TC](#) (2020, 2019), [IEEE CAL](#) (2026, 2022, 2021, 2020, 2018, 2017), [IEEE TMC](#) (2017), [IEEE Micro](#) (2022, 2020, 2019, 2018, 2017), [ACM TACO](#) (2026, 2019, 2018, 2016), [HCOMP](#) (2016), [IEEE ESL](#) (2015), [ACM TODAES](#) (2011), [ICS](#) (2018), [DAC](#) (2012, 2011)

#### Workshop and Tutorial Organizations

- Co-organizer: *Visual Computing for Computer Architects*, tutorials co-located with [ASPLOS 2023](#), [MICRO 2023](#), [ISCA 2024](#)
- Workshop and Tutorial Co-Chair, [MICRO 2022](#)
- Co-chair: Artifact Evaluation Committee, [IISWC 2022](#)
- Co-organizer: *RSS2: Workshop on Robustness and Safe Software 2.0 (RSS2)*, co-located with ASPLOS 2022, 2021
- Co-chair: *International Workshop on Performance Analysis of Machine Learning Systems (FastPath)*, co-located with ISPASS 2021
- Organizer: *Infrastructure and Methodology for SoC Performance and Power Modeling*, tutorials co-located with IISWC 2018, ASPLOS 2019, [ISCA 2019](#)
- Organizer: [Cognitive Edge Computing Workshop](#), co-located with MICRO 2017
- Co-chair: [Sensors to Cloud Architectures Workshop](#), co-located with HPCA 2017
- Web chair: [Cognitive Edge Computing Workshop](#), co-located with MICRO 2016
- Organizer: [MobiTools Workshop](#), co-located with ISCA 2016

**UNIVERSITY-  
WIDE  
SERVICE**

- *Committee on Undergraduate AI Degree Program, 2024-2025*
- *AI Ethics Committee, 2024-2025*
- *Policy Committee on International Collaborators, 2023-2024*

**DEPARTMENTAL  
SERVICE**

- *Faculty Recruitment Committee, 2024-2025*
- *PhD Admissions Committee, 2022-2023, 2021-2022, 2020-2021, 2019-2020, 2018-2019*
- *MS Admissions Committee in CS, 2021-2022*
- *MS Admissions Committee in GIDS, 2022-2023, 2021-2022, 2020-2021*
- *Undergraduate Curriculum Committee, 2021-2022, 2020-2021, 2019-2020*
- *Industrial Affiliates Exploration, 2020-2021*
- *Lab Committee, 2018-2019*
- *Colloquium Coordinator, 2018-2019*

**OUTREACH  
ACTIVITIES**

**University of Rochester Upward Bound Math and Science**

- Taught three-day workshop each summer to high-schoolers from the Vanguard Collegiate High School and Wilson High School
- Used Raspberry Pi as the platform to introduce programming to students, and incrementally built simple computer vision programs such as edge detectors.

**University of Rochester Pre-College Programs**

- Co-taught (with an undergraduate advisee Yongzhao Wu) an Introduction to Engineering course with a course project/demo on building an Arduino-based light sensor.

**Women in Engineering Program (WEP)**

- Speaker at high school-focused summer camps at UT Austin with an emphasis on inspiring high school female about engineering (~75 high school senior women)
- Mentored two female sophomores in the Graduates Linked with Undergraduates in Engineering ([GLUE](#)) Program.

**TEACHING  
EXPERIENCE**

**Instructor (University of Rochester)**

- Fall 2026, CSC 259/459 *Computer Imaging and Graphics*
- Spring 2026, CSC 252/452 *Computer Organization*
- Fall 2025, CSC 259/459 *Computer Imaging and Graphics*
- Spring 2025, CSC 252/452 *Computer Organization*
- Fall 2024, CSC 259/459 *Computer Imaging and Graphics*
- Fall 2024, CSC 412 *Introduction to Augmented and Virtual Reality* (co-instructor)
- Spring 2023, CSC 252/452 *Computer Organization*
- Fall 2022, CSC 292/572 *Mobile Visual Computing*
- Fall 2022, CSC 412 *Introduction to Augmented and Virtual Reality* (co-instructor)
- Spring 2022, CSC 252/452 *Computer Organization*
- Fall 2021, CSC 292/572 *Mobile Visual Computing*
- Fall 2021, CSC 412 *Introduction to Augmented and Virtual Reality* (co-instructor)

- Spring 2021, CSC 414 *Selected Topics on Augmented and Virtual Reality* (co-instructor)
- Spring 2021, CSC 252/452 *Computer Organization*
- Fall 2020, CSC 292/572 *Mobile Visual Computing*
- Fall 2020, CSC 412 *Introduction to Augmented and Virtual Reality* (co-instructor)
- Spring 2020, CSC 252/452 *Computer Organization*
- Spring 2019, CSC 252/452 *Computer Organization*
- Fall 2018, CSC 292/572 *Mobile Systems Architecture*
- Spring 2018, CSC 252/452 *Computer Organization*

#### Teaching Assistant (UT Austin)

- Spring 2013, *Dynamic Compilation*, with Vijay Janapa Reddi
- Spring 2011, *Computer Architecture*, with Yale N. Patt
- Fall 2010, *Introduction to Embedded Systems*, with Jonathan W. Valvano

#### MENTORING

#### University of Rochester

##### Current Ph.D. Students:

- Abhishek Tyagi
- Nisarg Ujjainkar
- Ethan Chen
- Weikai Lin
- Han Yan
- Sheng Zhao

##### Supervised Ph.D. Dissertations:

- Yu Feng, Ph.D. 2023. Dissertation: *Systematic Optimizations for Efficient Mobile Visual Computing*. Current position: Assistant Professor at Shanghai Jiaotong University
- Yiming Gan, Ph.D. 2023. Dissertation: *Improving Fault Tolerance of Computing Systems for Autonomous Machines*. Current position: Assistant Professor at Institute of Computing Technology, Chinese Academy of Sciences

##### MS Alumni:

- Raj Rajwade (M.S. 2023)
- Suumil Roy (M.S. 2023)
- Shreyan Goswami (M.S. 2022)
- Zeyi Pan (M.S. 2021)
- Hanlin Gao (M.S. 2021)
- Yi Yang (M.S. 2021)
- Ting Wu (M.S. 2020; now at Ebay)
- Tianqi Yang (M.S. 2020; now at Amazon)
- Tiancheng Xu (M.S. 2020; Ph.D. at Rice University)
- Chi-chun Chen (M.S. 2019; now at Cray Inc.)
- Boyuan Tian (M.S. 2019; Ph.D. at UIUC)

**Undergraduate Alumni:**

- Yongzhao Wu (Summer 2023)
- Ziyang Yuan (Summer 2023)
- Samuel Lee (Spring 2023)
- Zeyu Niu (Spring 2023)
- Ethan Shahan (Summer 2022–Spring 2023)
- Christopher Bruinsma (Spring 2022)
- Jiayi He (Spring 2022–Summer 2022; MS at CMU)
- Matan Kotler-Berkowitz (Spring 2022, Fall 2022, Spring 2023)
- Jennifer Yu (Summer 2022)
- Junhua Huang (Summer 2022)
- Meisen Hu (Summer 2022)
- Ruihan Xu (Summer 2022)
- Muhammad Qasim (Summer 2022)
- Shengyi Jia (Summer 2022; MS at John Hopkins University)
- Nikhil Khanna (Summer 2022)
- Gunnar Hammonds (Summer 2022)
- Mohamed ali manai (Summer 2022)
- Siddhant Choudhary (Summer 2022)
- Edmund Sepeku (Spring 2021)
- Kharissa King (Spring 2021)
- Elias Neuman-Donihue (Summer 2021–Spring 2022)
- Samuel Triest (Spring 2020–Spring 2021; MS/Ph.D. at CMU)
- Yawo Alphonse Siatitse (Spring 2020; MS at John Hopkins University)
- Qiuyue Sun (Summer 2018–Spring 2019; now at ByteDance)
- Sifan Ye (Spring 2019–Summer 2020; MS at Stanford University)
- Weituo Kong (Fall 2019; MS at Brown University)
- Oliver Zhang (Summer 2018; transferred to Univ. of Michigan)
- Noah Helterbrand (Summer 2018)
- Tolga Furkan Aktas (Fall 2019)
- Jessica Ervin (Fall 2019)

**Previously at UT Austin**

- Hannah Peeler, Undergraduate student, 2016
- Janna Tulabot, Undergraduate student, 2016
- Matthew Halpern, Ph.D. student, 2013 - 2017
- Wenzhi Cui, Ph.D. student, 2015 - 2017

## DISSERTATION COMMITTEE

### Ph.D.

- Carlos Mauricio Villegas Burgos (Optics, 2025): *Joint Optimization Framework of Metasurface-based Optics with Deep Learning Systems*
- Wuxiucheng Wang (ECE, 2025): *Integrated Silicon Photonic Optical Phased Array, Devices, Circuits and Systems, for Free-Space Optical Communication*
- Tianrui Ma (ECE, Washington University in St. Louis, 2024): *Efficient Data-Driven Machine Vision: A Co-Design of Circuit, Algorithm, and Architecture for Edge Vision Sensors*
- Eslam Elmitwalli (ECE, 2024): *Applications of Randomness in Hardware Security and Combinatorial Optimization*
- Irving Barron (ECE, 2024): *Design and Applications of 2D Barcodes Using Dual Modulation*
- Yuanyuan Pan (Math, 2024): *The Damped Wave Equation and Associated Polymer*
- Uday Kumar Redd Vengalam (ECE, 2023): *Extending Ising Machines for solving Machine learning problems and LDPC codes*
- Narges Mohammadi (ECE, 2024): *Integrating Learning-based and Physical Model-based Methods for Elasticity Reconstruction*
- David Lippman (Optics, 2023): *Freeform gradient-index optical design and metrology for reduced system size and weight*
- Sayak Chakraborti (CS, 2022): *Opportunistic Resource Management: Resource Utilization in Datacenters*
- Divya Ojha (CS, 2022): *Defending Against Microarchitectural Side Channel Leaks*
- Haichuan Yang (CS, 2020): *Sparse Learning for Model Optimization*
- Daniel Nikolov (Optics, 2020): *Software and Hardware Enabling the Next-Generation Near-Eye Displays*
- Kan Xu (ECE, 2020): *Power Delivery in High Current 3-D Systems*
- Hoda Sadat Ayatollahi Tabatabaei (ECE, 2018): *Energy Balancing in Wireless Networks with MIMO Communications*

### M.S.

- Junfu Zheng (Optics, 2023): *Measurement of gradient-index materials with transmission deflectometry*
- Yiwen Fan (Optics, 2020): *Numerical Calculation of Zernike Polynomials and the Sample Selection Method of NURBS Spline Generation*

### B.S.

- Ezra Tock (CS, 2025): *Analysis of N-BVH Ray Tracing for Remote Use and for Extension to Dynamic Scenes*
- Aayush Poudel (CS, 2023): *Compressing Ray Trajectory Mapping using Bézier Curves*
- Elias Neuman-Donihue (CS, 2022): *Fast Rendering of Massive Point Clouds*
- Benned Hedegaard (CS, 2022): *Sign-Informed Semantic Mapping for Language Interaction*
- Sifan Ye (CS, 2020): *3D Reconstruction from Colored Point Clouds with Detail Transfer*
- Samuel Triest (CS, 2020): *Unsupervised Reinforcement Learning in Environments with Strong Priors*